Fiscal Year:	FY 2020	**Task Last Updated:**	FY 05/29/2020
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Project Title:	Integrative Personalized Omics Profiling Next Steps: Detection and Classification of Deviations from Wellness		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 TRA BRASH1801: Translational Research Institute for Space Health (TRISH) Biomedical Research Advances for Space Health
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Piermarocchi, Carlo Ph.D. (Michigan State University)		
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Task Description:	Precision medicine is an emergent interdisciplinary field that utilizes state-of-the art technologies to detect changes in individual wellness, diagnose, treat, and possibly prevent diseases. Previous studies, by us and others, have focused on establishing the feasibility of personalized precision medicine, by measuring multiple time signals in individuals (generalized omics). These signals typically include medically relevant measures, such as clinical tests, blood panels, gene expression, and data from personal health devices. However, algorithmic development and sophisticated methods to analyze, classify, and biologically annotate such omics are still underdeveloped. In our proposed investigation, we will focus on the essential next steps to implement such integrative Personal Omics Profiling (iPOP) on individual astronauts, and detect and classify adverse medical events through the analysis of generalized omics signals. Objectives: Our objectives are to: (i) Generate new computational methodology to establish the baseline healthy characteristics in individual personnel, and detect baseline deviations that predict the onset of any adverse medical event. (ii) Classify departures from individualized wellness according to collective biological/physiological signal characteristics and behavior. We will annotate the classes based on medical relevance, for instance by associating groups of signals to a specific disease. Our methodology will compare medical baseline deviations both within and across individuals. Methods: To achieve our objectives we plan to develop sophisticated computational methods in statistical time series analysis and dynamic network theory to characterize and categorize different biological/physiological signals from individuals. We will develop and implement new algorithms that establish an individual's health baseline through the integration of thousands of signals, and that can detect deviations from a person's own baseline. The time series will be analyzed in terms of their underlying frequencies and limitations due to missing data or uneven sampling will be addressed. We will create a geometric representation of the signals, including networks that represent biological interactions between the omics that will allow us to both detect anomalies, but also infer the medical relevance of abnormal medical events. Significance to solicitation's objectives: Our proposed investigation is completely aligned with Topic 1 of the solicitation, to develop Predictive algorithms of health, behavior, and medical events. The frameworks and methods of this investigation are designed to be directly applied to the monitoring of spaceflight personnel. Deliverables of this project include: (i) novel computational methodology, (ii) open source software packages (including a standalone analysis tool), and (iii) a report assessing the predictive utility of different data types and tissue samples that may be used in flight (including saliva). Significance to NASA's interests and programs: Personalized monitoring of astronauts is already being performed, and through implementing our novel methods we will be able to improve NASA's ability to detect baseline abnormalities in flight, intervene with corrective measures, and maintain individualized wellness for personnel, which is mission critical.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We are developing predictive algorithms of health, behavior, and medical events. Our method is extensible and can be used to incorporate any dynamic data that can be measured during spaceflight -- including different sampling rates. Furthermore, our approach can be implemented on Earth as well using personnel medical monitors to detect medical events. The frameworks and methods of this investigation are being designed to be directly applied to the monitoring of spaceflight personnel. Our trend detection algorithm addresses multiple challenges in analyzing signal data from human subjects and in-mission personnel. Specifically in our algorithm, and the Python package PyIOmica which has been released (as outlined above), the following challenges are addressed: (i) the ability to integrate thousands of time series per person, (ii) carry out different normalizations per signal depending on the signal origin (e.g., transcripts versus metabolites), (iii) address the different time frames over which measurements may be taken (e.g., seconds versus hourly or daily measurements), (iv) have the ability to use many or few time points in the analysis, (v) incorporate methods to address missing data, (vi) address uneven sampling of signals (typically the case for in-flight or in-the-field measurements), (vii) integrate multiple layer information (from different measurements, using different monitors), (viii) provide utilities for visualization and facilitating the interpretation of data by non-specialists (including pathway annotations for gene-based or metabolite-based measurements). Deliverables of this project include: (i) novel computational methodology, (ii) open source software packages (including a standalone analysis tool), and (iii) a report assessing the predictive utility of different data types and tissue samples that may be used in flight (including saliva). Significance to NASA's interests and programs: Personalized monitoring of astronauts is already being performed, and through implementing our novel methods we will be able to improve NASA's ability to detect baseline abnormalities in flight, intervene early with corrective measures, and maintain individualized wellness for personnel, which is mission critical.
	[Ed. note May 2020: Report submitted by TRISH to Task Book in March 2020; covers reporting as of December 2019.] Precision medicine is an emergent interdisciplinary field that utilizes state-of-the art technologies to detect changes in individual wellness, diagnose, treat, and possibly prevent diseases. Previous studies, by us and others, have focused on establishing the feasibility of personalized precision medicine, by measuring multiple time signals in individuals (generalized omics). These signals typically include medically relevant measures, such as clinical tests, blood panels, gene expression, and data from personal health devices. However, algorithmic development and sophisticated methods to analyze, classify, and biologically annotate such omics are still underdeveloped. In our proposed investigation, we will focus on the essential next steps to implement such integrative Personal Omics Profiling (iPOP) on individual astronauts, and detect and classify adverse medical events through the analysis of generalized omics signals. Objectives: Our objectives are to: (i) Generate new computational methodology to establish the baseline healthy characteristics in individual personnel, and detect baseline deviations that predict the onset of any adverse medical event. (ii) Classify departures from individualized wellness according to collective biological/physiological signal characteristics and behavior. We will annotate the classes based on medical relevance, for instance by associating groups of signals to a specific disease. Our methodology will compare medical baseline deviations both within and across individuals. Methods: To achieve our objectives we plan to develop sophisticated computational methods in statistical time series analysis and dynamic network theory to characterize and categorize different biological/physiological signals from individuals. We will develop and implement new algorithms that establish an individual's health baseline through the integration of thousands of signals, and that can detect deviations from a person's own baseline. The time series will be

Task Progress:	analyzed in terms of their underlying frequencies and limitations due to missing data or uneven sampling will be addressed. We will create a geometric representation of the signals, including networks that represent biological interactions between the omics that will allow us to both detect anomalies, but also infer the medical relevance of abnormal medical events. Significance to solicitation's objectives: Our proposed investigation is completely aligned with Topic 1 of the solicitation, to develop Predictive algorithms of health, behavior, and medical events. The frameworks and methods of this investigation are designed to be directly applied to the monitoring of spaceflight personnel. Deliverables of this project include: (i) novel computational methodology, (ii) open source software packages (including a standalone analysis tool), and (iii) a report assessing the predictive utility of different data types and tissue samples that may be used in flight (including saliva). Significance to NASA's interests and programs: Personalized monitoring of astronauts is already being performed, and through implementing our novel methods we will be able to improve NASA's ability to detect baseline abnormalities in flight, intervene with corrective measures, and maintain individualized wellness for personnel, which is mission critical. Key Findings (Year 1): We have completed objective (i): We have developed an algorithm for adverse event detection that can characterize baseline deviations from any time series signal. We have implemented the algorithm in the published Python package PyIOmica. The algorithm can integrate thousands of time series with different normalizations and time frames, and can address issues of missing data and uneven sampling. For measurements (time-series) with gene-annotated information, and/or metabolite data, pertinent pathways that are affected can be identified to assess biological and medical significance. Furthermore the algorithm can be used to incorporate any time-signal from different medical monitors used in-flight. The algorithm will act as a backend base for the deliverable AstroIO application that will integrate multimodal information from various health monitors in Year 2. We have implemented the methodology leading to Aim 1 to classify multi-omics measurements in saliva from a single individual as a proof of principle for saliva utility in non-invasive diagnostic monitoring of physiological state and adverse events (hourly and daily monitoring using saliva RNA/protein and extracellular vesicles). Research Plan (Year 2): We plan to complete objective (ii). We will compare individualized wellness departures, and annotate different adverse medical events using retrospective astronaut data from Lifetime Surveillance of Astronaut Health (LSAH) (data request approved in Year 1). This will allow us to characterize and obtain signatures of adverse medical events, and improve early detection and potential timely treatment. We will integrate multiple layers of time series signals from different measurements and devices to build networks of biological interactions,that can assess medical significance of events. The algorithms will be integrated in the deliverable open source desktop application AstroIO. Retrospective astronaut data will be used to validate all algorithms and for categorizing already observed medical events. Finally, we will complete our report on the utility of saliva for non-invasive in-flight diagnostics.
Bibliography Type:	Description: (Last Updated: 10/30/2023)
Articles in Peer-reviewed Journals	Domanskyi S, Piermarocchi C, Mias GI. "PyIOmica: Longitudinal omics analysis and trend identification." Bioinformatics. 2019 Nov 28 Epub ahead of print. https://doi.org/10.1093/bioinformatics/btz896 ; PMID: 31778155; PMCID: PMC7141865 , Nov-2019
Articles in Peer-reviewed Journals	Mias GI, Zheng M. "The MathIOmica Toolbox: General analysis utilities for dynamic omics datasets." Curr Protoc Bioinformatics. First published:18 December 2019. https://doi.org/10.1002/cpbi.91 ; PMID: 31851777 , Dec-2019